

Day: 21

* Why Filters Matter

- A PCAP can have millions of packets. Without filters you are blind. Filters are the difference between a 10-hour investigation and a 10-minute one. Every experienced analyst has a personal cheatsheet they use every single day.

* Display Filter vs Capture Filter

Type	When Applied	Syntax	Use
Capture filter	During live capture	BPF Syntax	Limit what gets captured
Display filter	After capture	Wireshark Syntax	Show/hide packets

- Always use Display filters for analysis — they are non-destructive (original PCAP unchanged).

* Filter Syntax Rules

- Equals: `== ip.src == 192.168.1.1`
- Not equals: `!= ip.src != 192.168.1.1`
- Greater than: `> ip.ttl > 64`

- Less than: $<$ ip.ttl < 60
- Contains string: contains http.host contains "google"
- Matches regex: matches dns.qry.name matches ".*\.\ru\$"
- Not: ! !dns
- AND: && tcp && ip.src == 10.0.0.1
- OR: || tcp || udp

* The Master Cheatsheet

1) Basic Protocol Filters:

tcp → All TCP traffic

udp → All UDP traffic

dns → All DNS traffic

http → All HTTP traffic

tls → All TLS/HTTPS traffic

arp → All ARP traffic

icmp → All ICMP traffic

smtp → All SMTP email traffic

ftp → All FTP traffic

ssh → All SSH traffic

2) IP and Port Filters:

ip.src == x.x.x.x → Traffic FROM specific IP

ip.dst == x.x.x.x → Traffic TO specific IP

ip.addr == x.x.x.x → Traffic To or FROM specific IP

tcp.port == 4444 → Traffic on port 4444

tcp.dport == 443 → Traffic TO port 443

tcp.sport == 80 → Traffic FROM port 80

ip.ttl < 10 → ~~Has~~ Unusually low TTL
— Suspicious

ip frag offset > 0 → Fragmented packets

3) Malware Hunting Filters:

tcp.flags.syn == 1 → SYN packets (Used for port scanning, new connection)

tcp.flags.syn == 1 → SYN only (No ack)
&& tcp.flags.ack == 0 (Used for port scan)

tcp.flags.reset == 1 → RST packets (Rejected connections)

`tcp.port == 1337` → Common malware port
(C2 detection)

`http.request.method == "POST"` → HTTP POST requests
(Data exfiltration)

`http.request.method == "GET"` → HTTP GET requests
(file downloads)

frame contains "password" → packets with "password"
(Credential theft)

frame contains "cmd.exe" → packets with cmd.exe
(Malware execution)

frame contains ".exe" → Exe file references
(Malware download)

4) DNS Filters:

`dns` → All DNS

`dns.qry.type == 1` → A record queries only
(IPv4 lookups)

`dns.flags.rcode == 3` → NXDOMAIN responses
(DNS malware failures)

`dns.qry.name contains ".ru"` → .ru domains
(Often malicious)

`dns.qry.name.len > 50` → Long domain names
(DNS tunneling)

`dns.count.answers == 0` → queries with no answer (Failed lookups)

5) TLS/HTTPS Filters:

`tls` → All TLS traffic (HTTPS Overview)

`tls.handshake.type == 1` → Client Hello only (See SNI)

`tls.handshake.extensions.server_name` → SNI field (e.g. domain in encrypted traffic)

`tls.record.version == 0x0300` → SSL 3.0 (old/weak) (vulnerable encrypt)

6) TCP Analysis Filters:

`tcp.analysis.retransmission` → Retransmitted packets (Connection problems)

`tcp.analysis.duplicate_ack` → Duplicate Acks (Congestion/Issues)

`tcp.analysis.zero_window` → Zero window (Overwhelmed receiver)

`tcp.stream eq N` → Specific TCP stream N (follow conversation)

`tcp.len > 0` → Packets with data (Exclude empty Acks)

⇒ ARP + DHCP Filters:

arp → All ARP (Layer 2 overview)

arp.opcode == 1 → ARP requests (who is looking?)

arp.opcode == 2 → ARP replies (who is answering?)

arp.duplicate - address + detected → Duplicate MACs (ARP poisoning!)

bootp → All DHCP (IP assignment)

bootp.option.dhcp == 1 → DHCP Discover (New device joining)

bootp.option.dhcp == 2 → DHCP Offer (check for rogue server)

8) Content Search Filters:

frame contains "password" → Packets containing "password"

frame contains "admin" → Packet containing "admin"

frame contains "gate.php" → Common C2 URL

frame contains "cmd.exe" → Command execution

http.uri contains "gate" → HTTP URI search

http.user_agent contains "curl" → curl user agent

http.host contains "pastebin" → Pastebin
(common for malware)